

Exploring the Impact of Lifestyle on Productivity



Data Analysis in Software Engineering 25/26

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Objective and Approach

How do everyday lifestyle factors affect productivity?
The project explores this through three complementary analyses, combining pattern discovery (clustering) with prediction (regression).

Factors examined include stress, diet quality, screen time, sleep, exercise, and restorative habits such as coffee versus power naps.

SHARED PIPELINE

Data Preprocessing:

missing values, column standardization, duplicate rows

Data Analysis:

outlier analysis, correlations, distributions

Clustering Analysis:

K-Means + DBSCAN

Predictive Modeling:

Decision Tree + Random Forest

Project structure

01

Mental Health and Productivity Analysis

Studies the effects of long-term lifestyle habits on productivity

20,000 rows and **8** features

02

Coffee vs Power Nap Analysis

Studies the effects of interventions on productivity

1,000 rows and **11** features

03

Merged Datasets

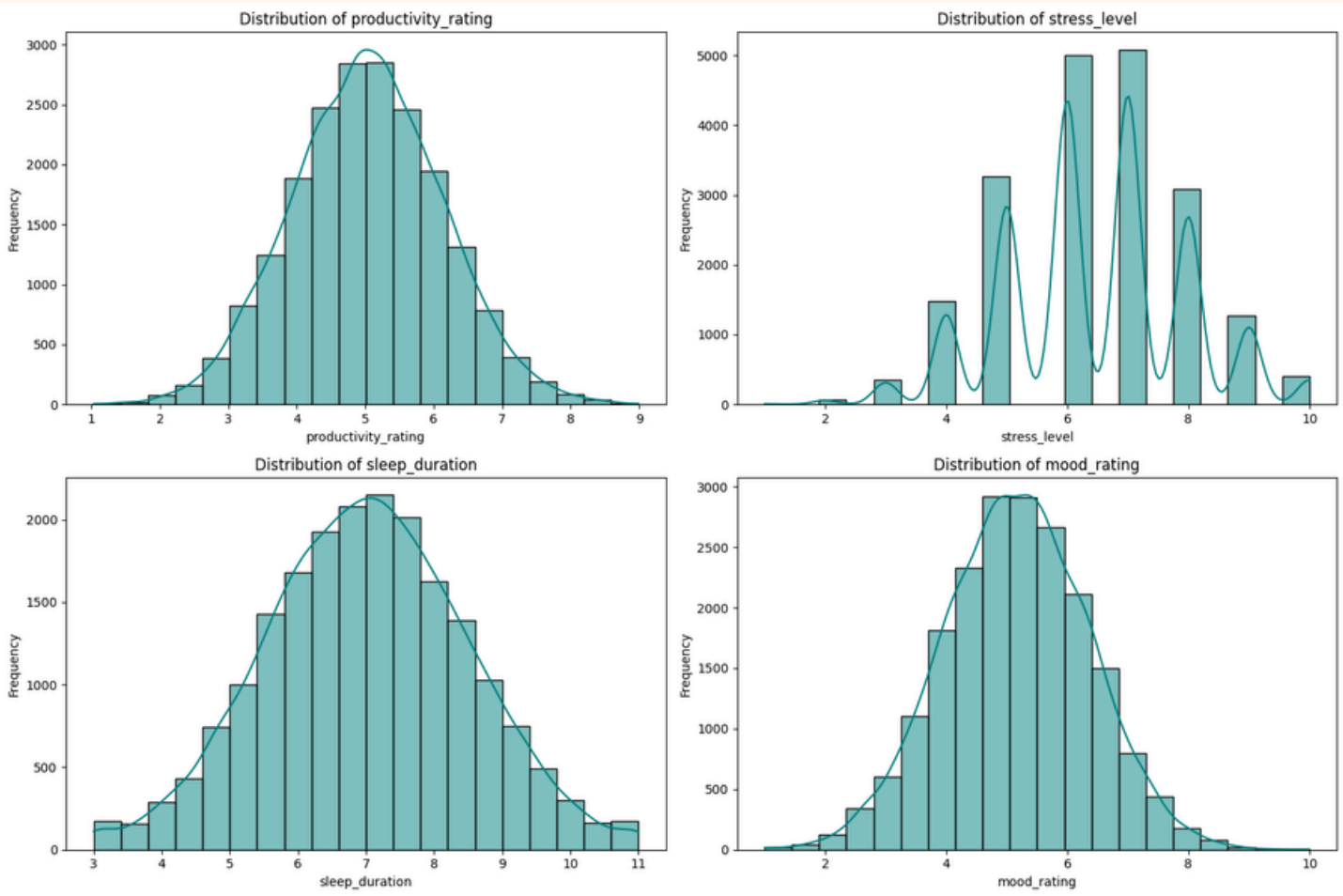
Merged with K-Nearest Neighbors

1,000 rows and **15** features

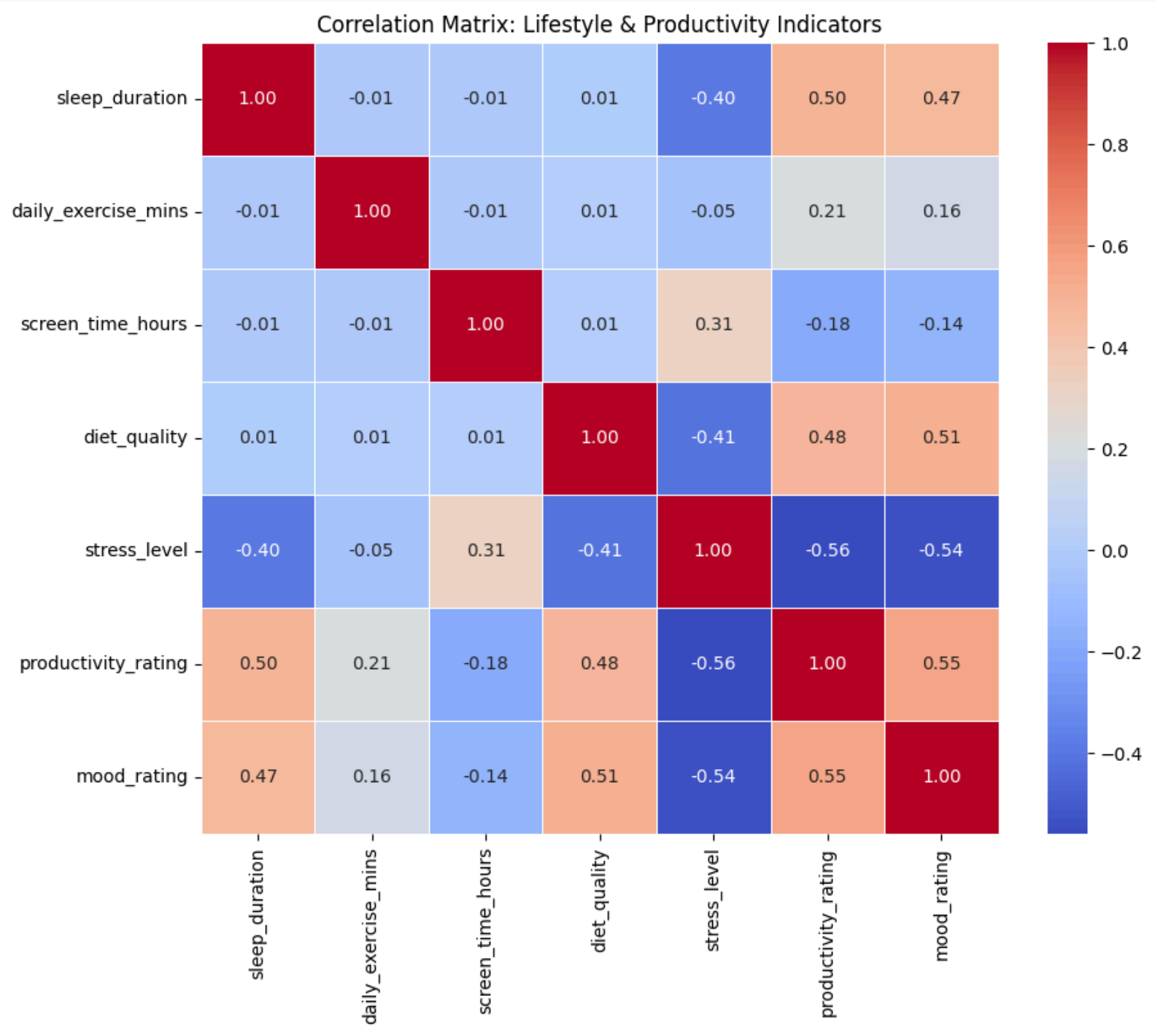
1. Mental Health & Productivity Analysis

Data Analysis

We also did outlier analysis, relationships scatter plots.



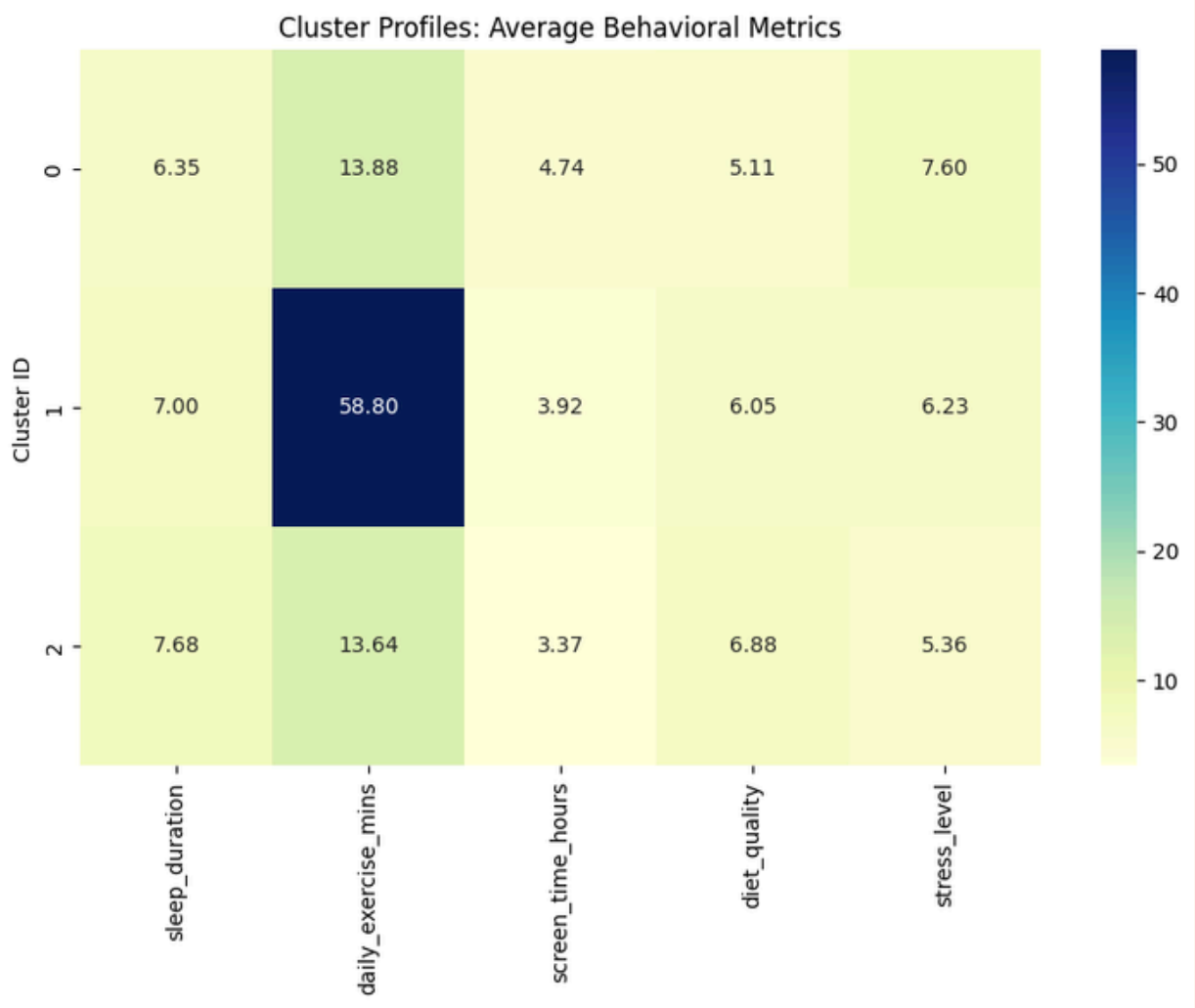
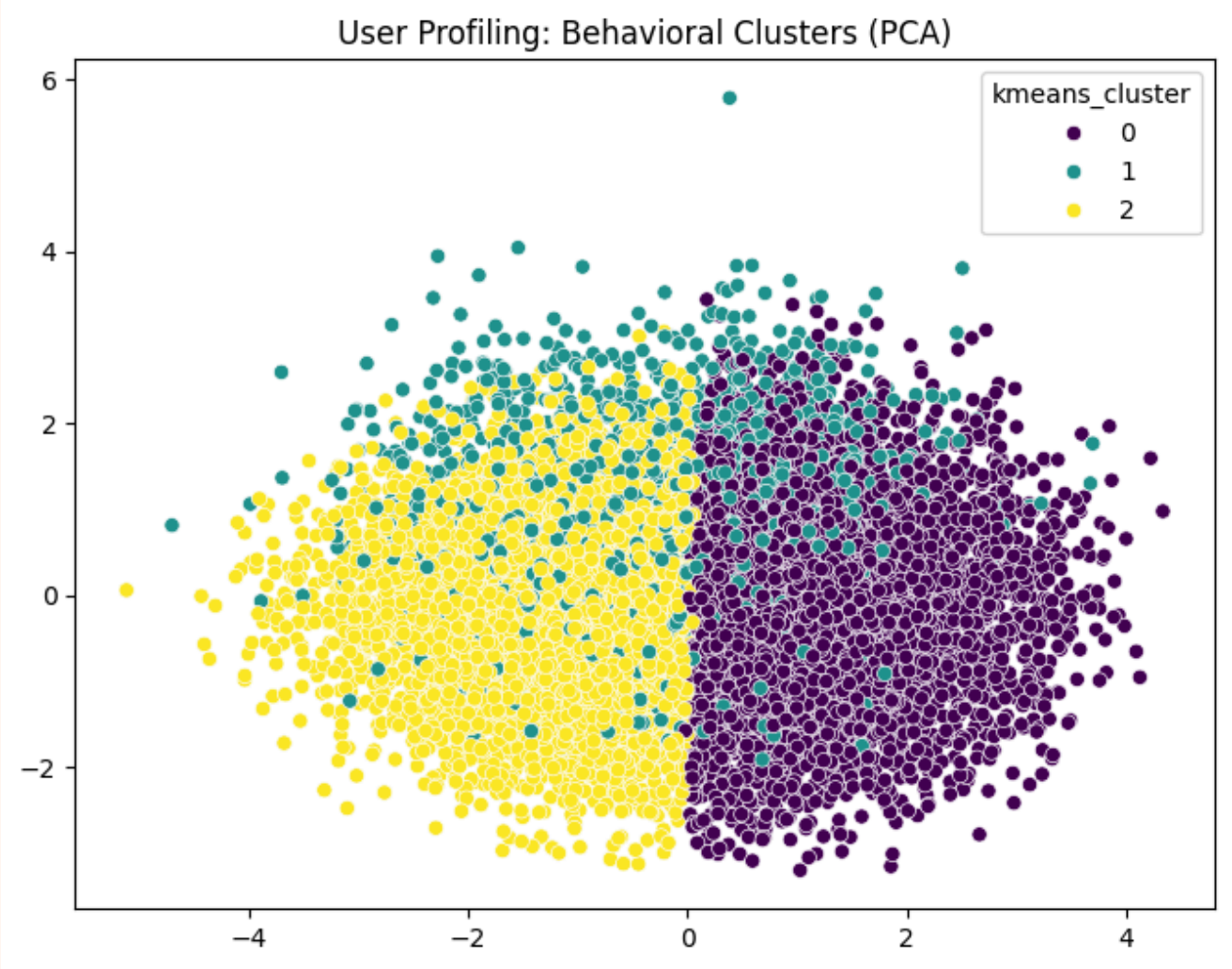
Most variables follow a roughly centered distribution, while stress shows greater spread and variability.



The correlation matrix shows that productivity is positively associated with sleep duration, diet quality, and mood, while stress has the strongest negative relationship with productivity.

1. Mental Health & Productivity Analysis

Clustering Analysis K-Means



We also tested DBSCAN, but it did not produce meaningful clusters beyond one dominant group.

1. Mental Health & Productivity Analysis

Predictive Modelling

Without feature engineering

— Cross-Validated R^2 (5-fold, mean $\pm$ std) —
DummyRegressor (mean): -0.000 ± 0.000
DecisionTree: 0.490 ± 0.006
RandomForest: 0.515 ± 0.009

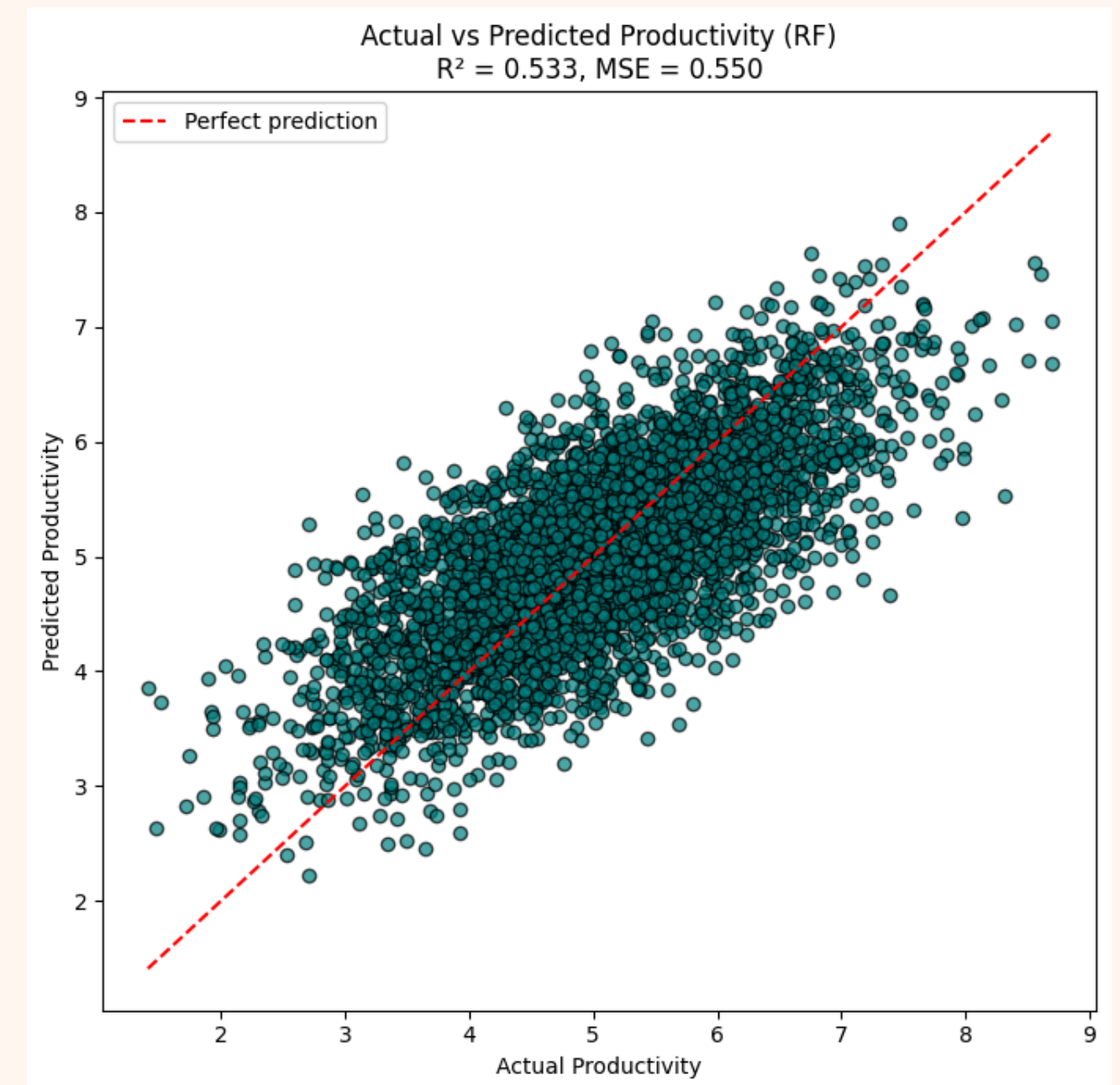
— Holdout Metrics (20% test split) —
DummyRegressor R^2 : -0.000 | MSE: 1.177
DecisionTree R^2 : 0.485 | MSE: 0.606
RandomForest R^2 : 0.510 | MSE: 0.577

With feature engineering

— Cross-Validated R^2 (5-fold, mean $\pm$ std) —
DummyRegressor (mean): -0.000 ± 0.000
DecisionTree: 0.522 ± 0.010
RandomForest: 0.535 ± 0.007

— Holdout Metrics (20% test split) —
DummyRegressor R^2 : -0.000 | MSE: 1.177
DecisionTree R^2 : 0.516 | MSE: 0.570
RandomForest R^2 : 0.533 | MSE: 0.550

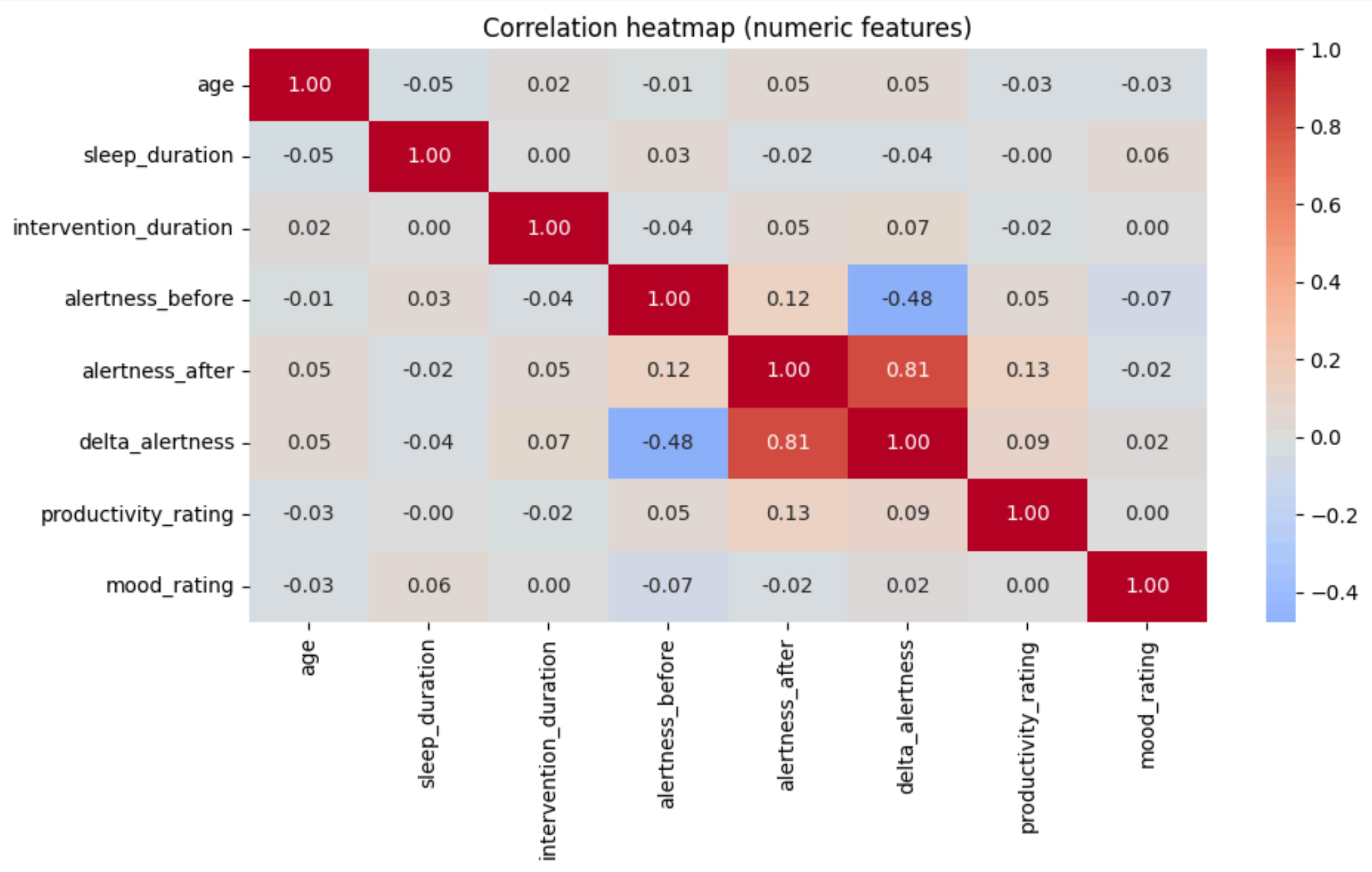
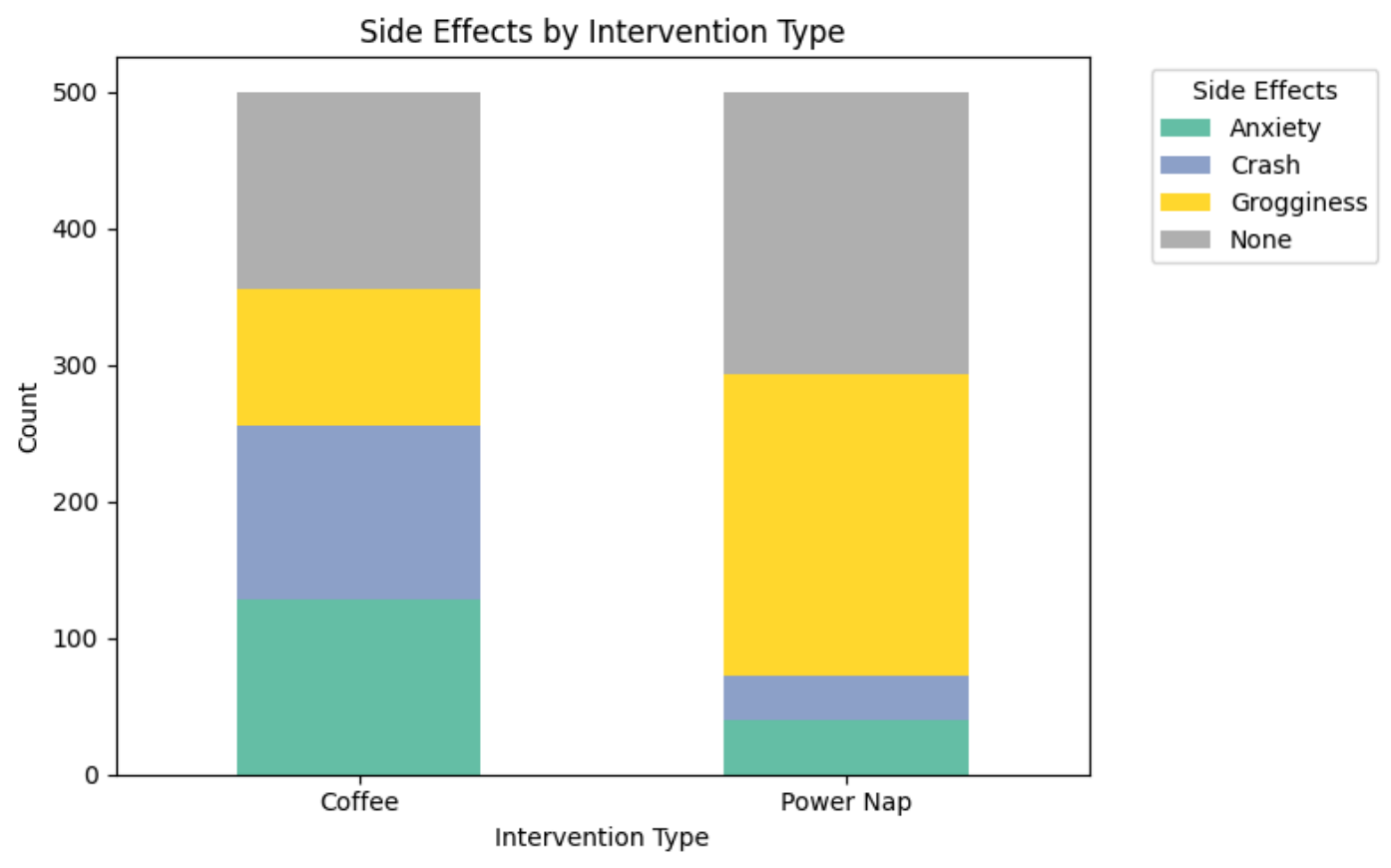
Feature engineering produced a small but consistent improvement in predictive performance, increasing Random Forest R^2 from about 0.51 to 0.53 and confirming it as the best-performing model, while the dummy baseline showed no predictive value.



2. Coffee vs Power Nap Analysis

Data Analysis

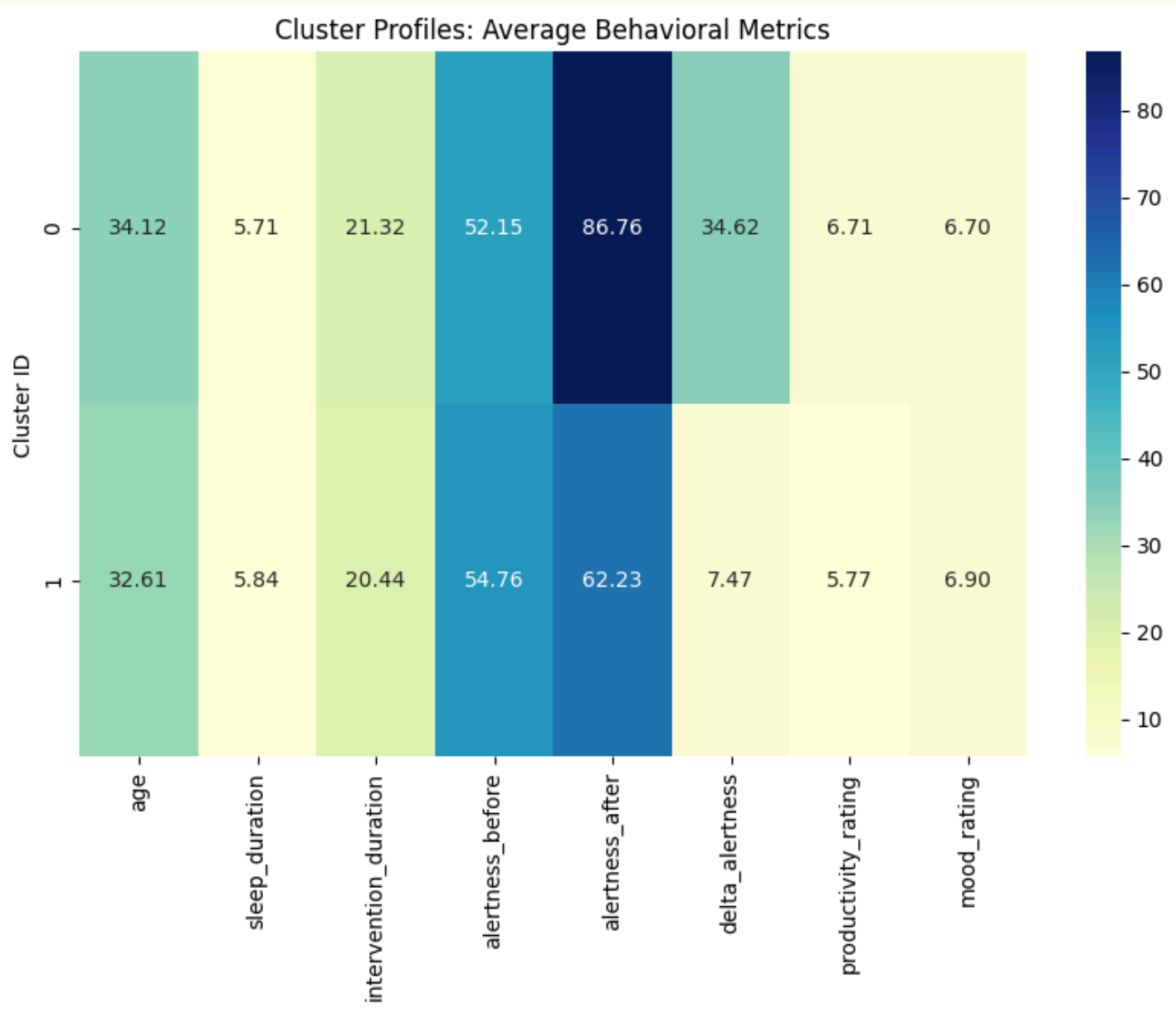
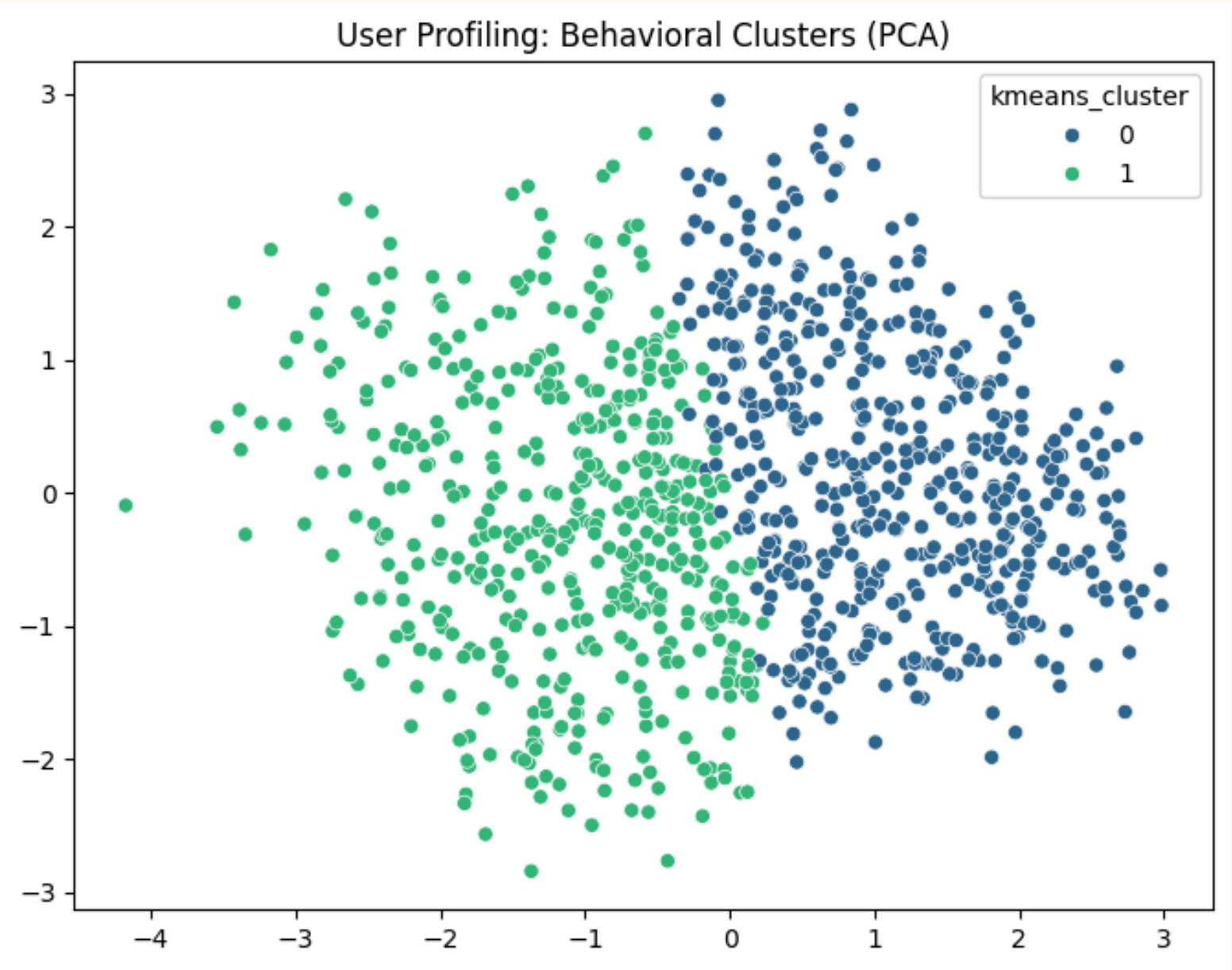
We also did outlier analysis, data visualization per occupation and age, comparative distribution analysis.



The correlation heatmap shows that most variables have weak relationships with productivity, with the strongest association appearing between alertness after the intervention and change in alertness.

2. Coffee vs Power Nap Analysis

Clustering Analysis K-Means



We also tested DBSCAN, but it did not produce meaningful clusters beyond one dominant group.

2. Coffee vs Power Nap Analysis

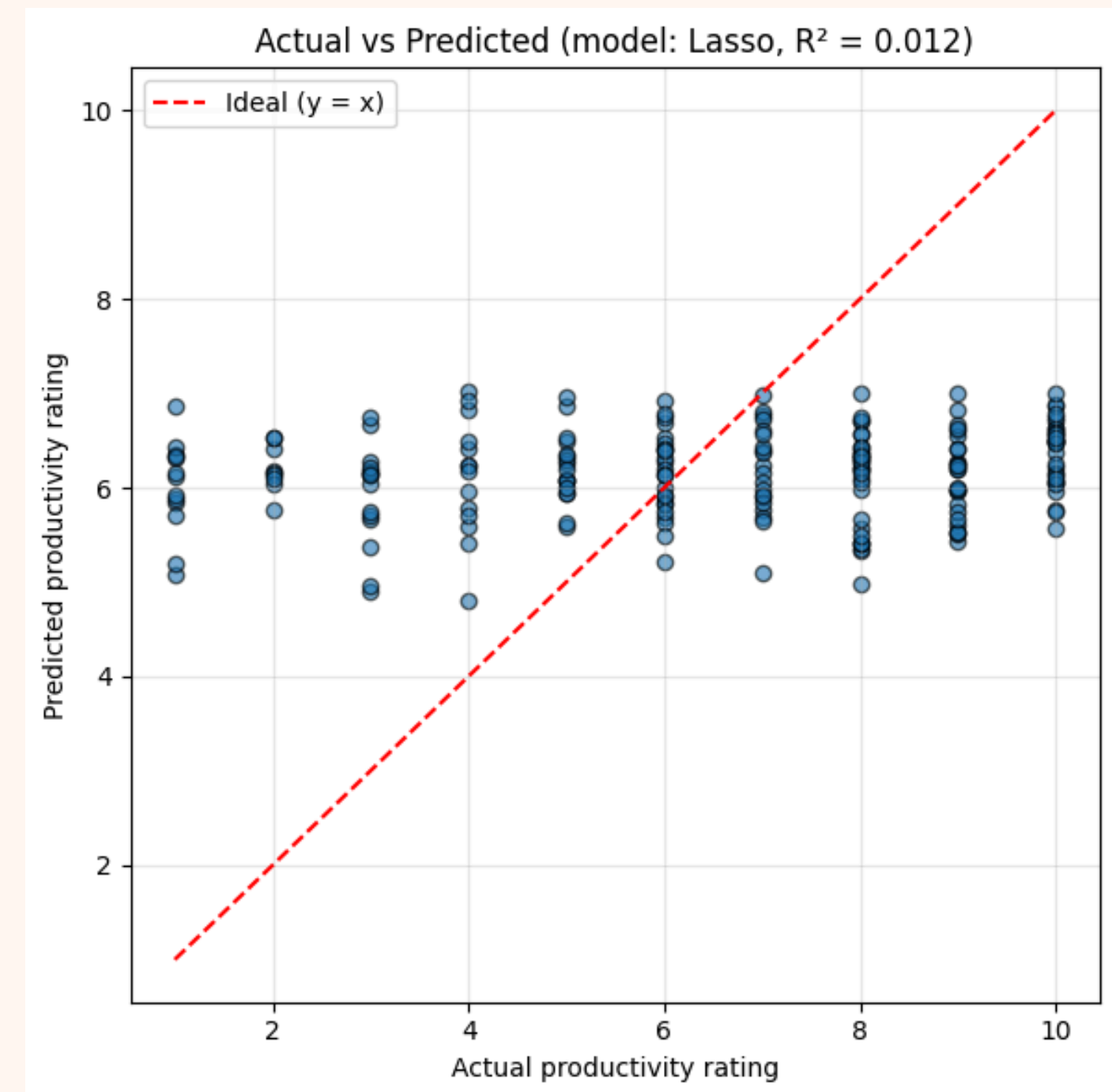
Predictive Modelling

6 approaches were tested, this being the best one

— Cross-Validated R^2 (5-fold, mean $\pm$ std) —
DummyRegressor: -0.003 ± 0.002
Ridge: 0.004 ± 0.017
Lasso: 0.005 ± 0.020
ShallowTree: -0.058 ± 0.032

— Holdout Metrics (20% test split) —
DummyRegressor: $R^2 = -0.009$ | MSE = 7.552
Ridge: $R^2 = 0.012$ | MSE = 7.399
Lasso: $R^2 = 0.012$ | MSE = 7.400
ShallowTree: $R^2 = -0.006$ | MSE = 7.533

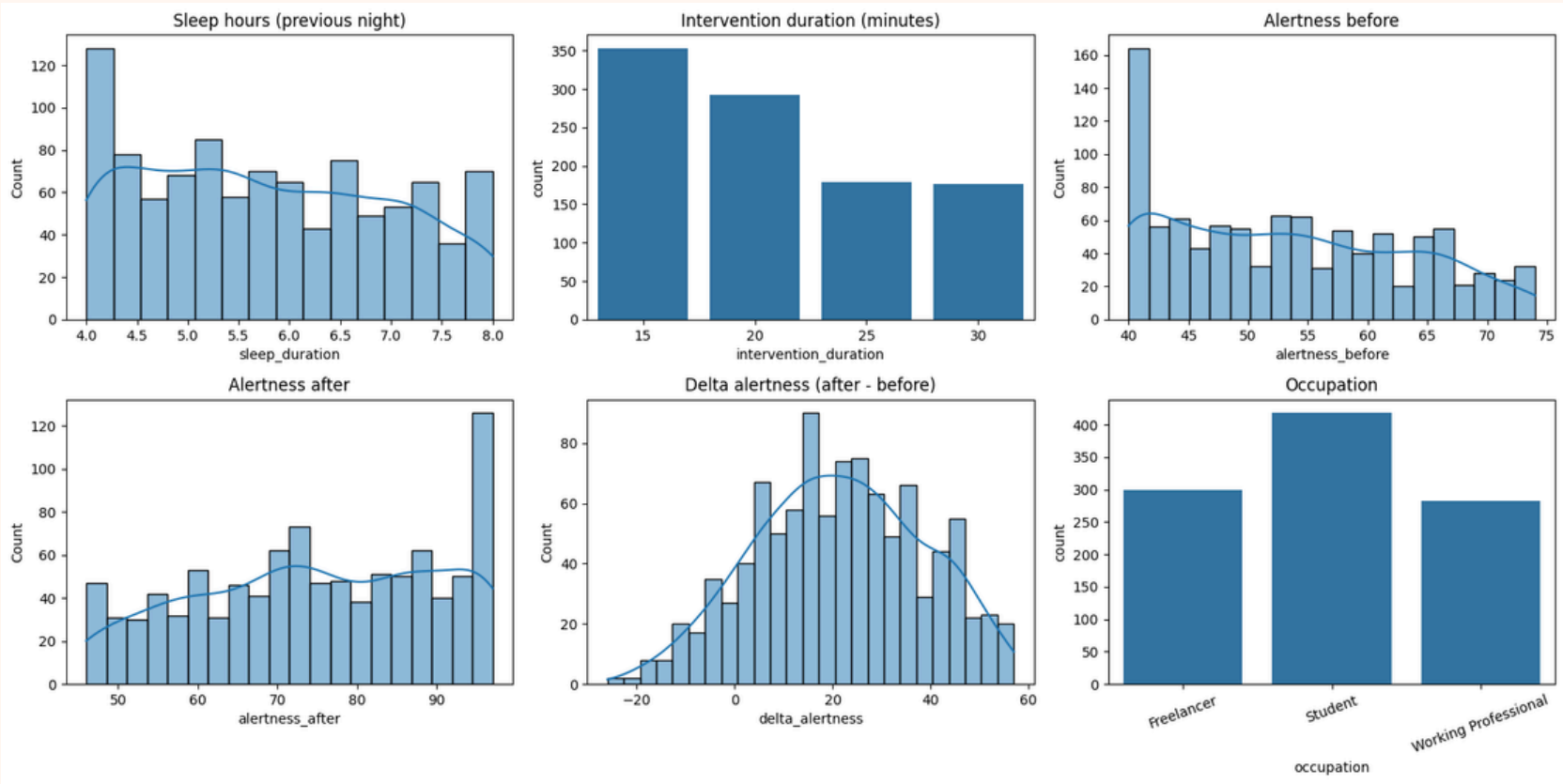
The very low $R^2 \approx 0.01$ indicates that productivity was largely unpredictable from the nap vs. coffee variables, likely because the sample was small, other important factors were not measured, and the productivity ratings themselves may have been noisy.



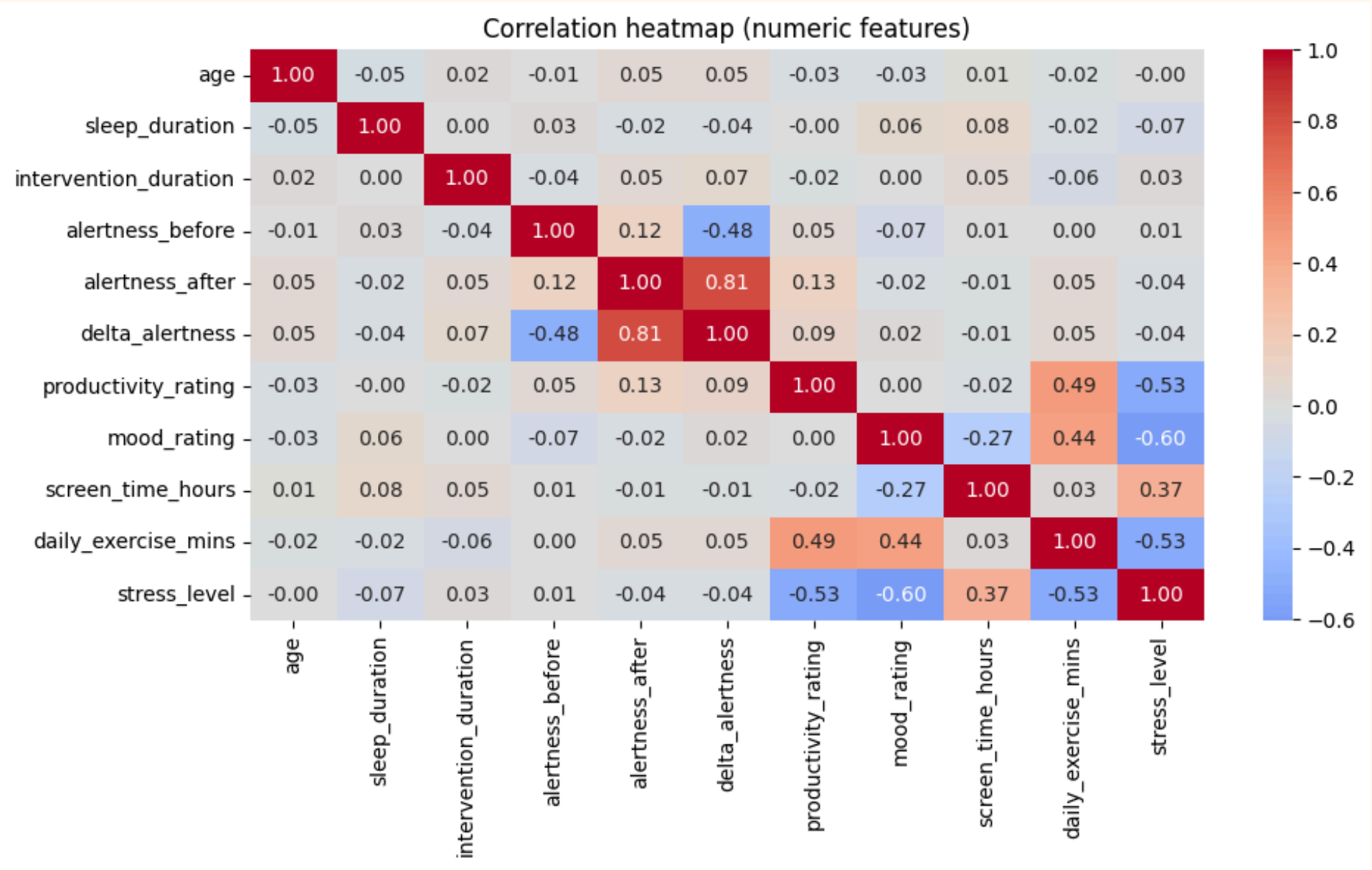
3. Merged Analysis

Data Analysis

We also did outlier analysis, alertness outcome analysis, group-level performance comparison, analysis if side effects.



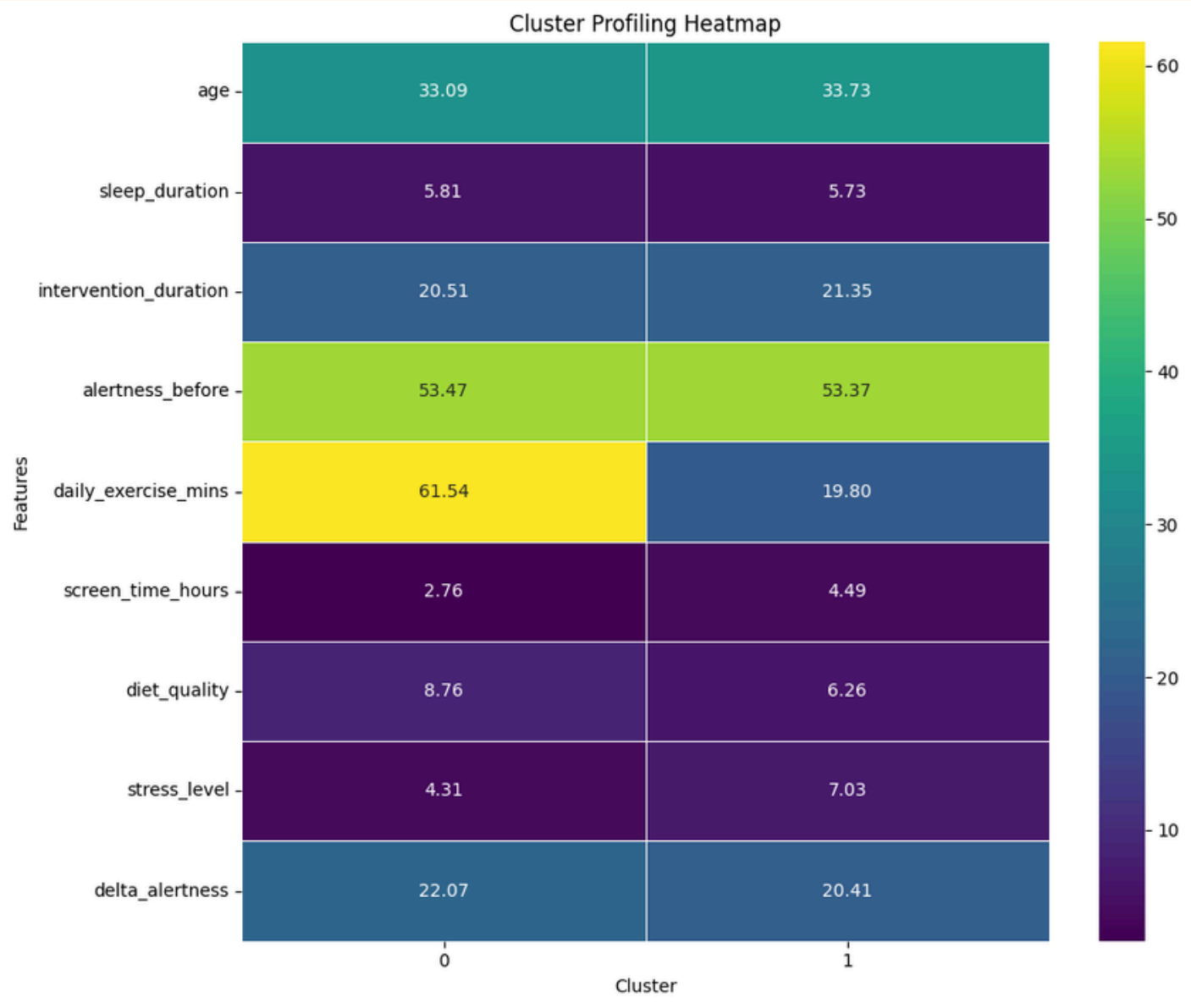
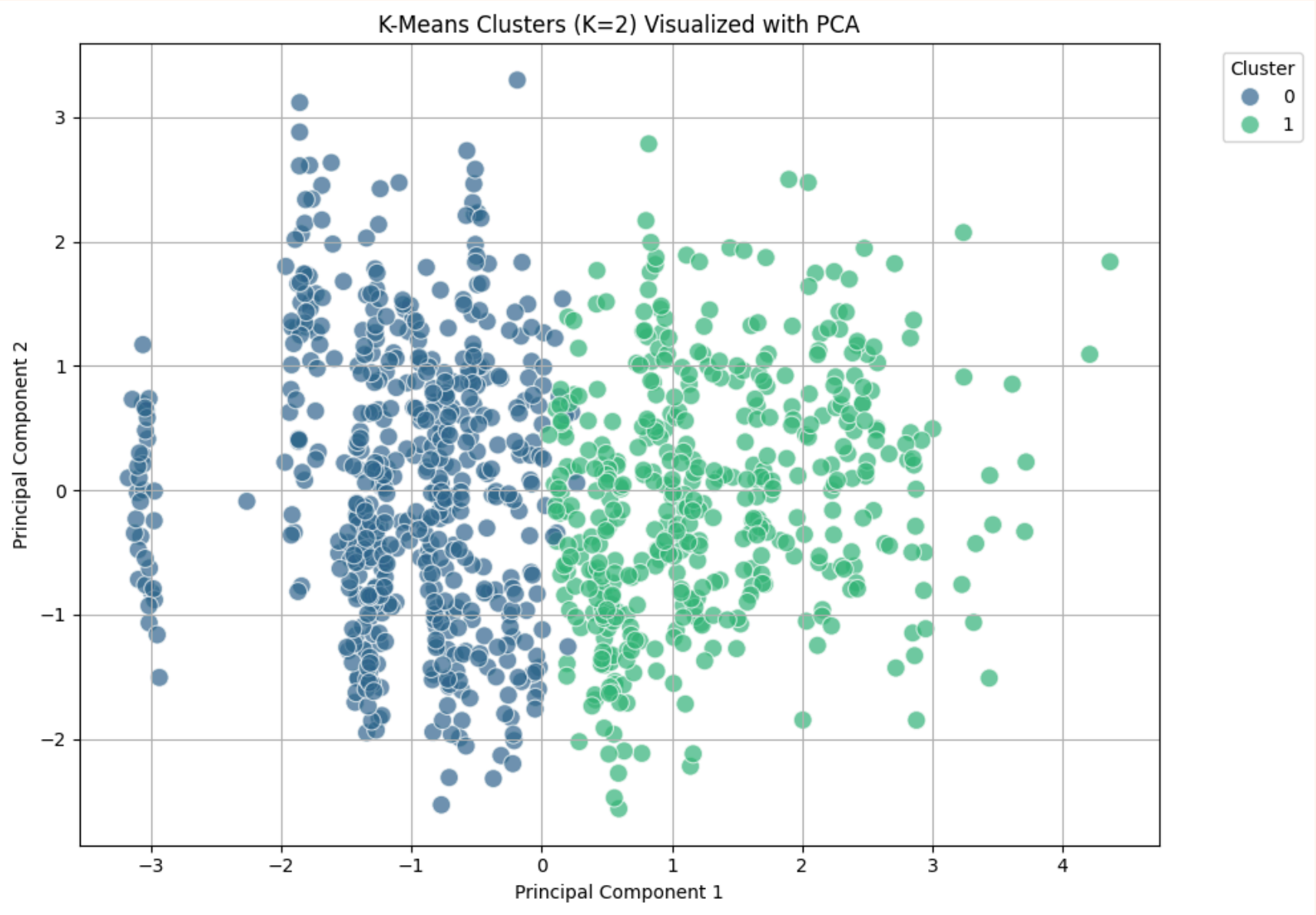
The distributions show varied sleep and intervention patterns, but alertness generally increased after the intervention, with most participants showing a positive change.



The correlation heatmap shows that productivity is most strongly linked to stress, exercise, and mood, while age, sleep, and intervention duration show much weaker relationships in this dataset.

3. Merged Analysis

Clustering Analysis K-Means



We also tested DBSCAN, but it did not produce meaningful clusters beyond one dominant group.

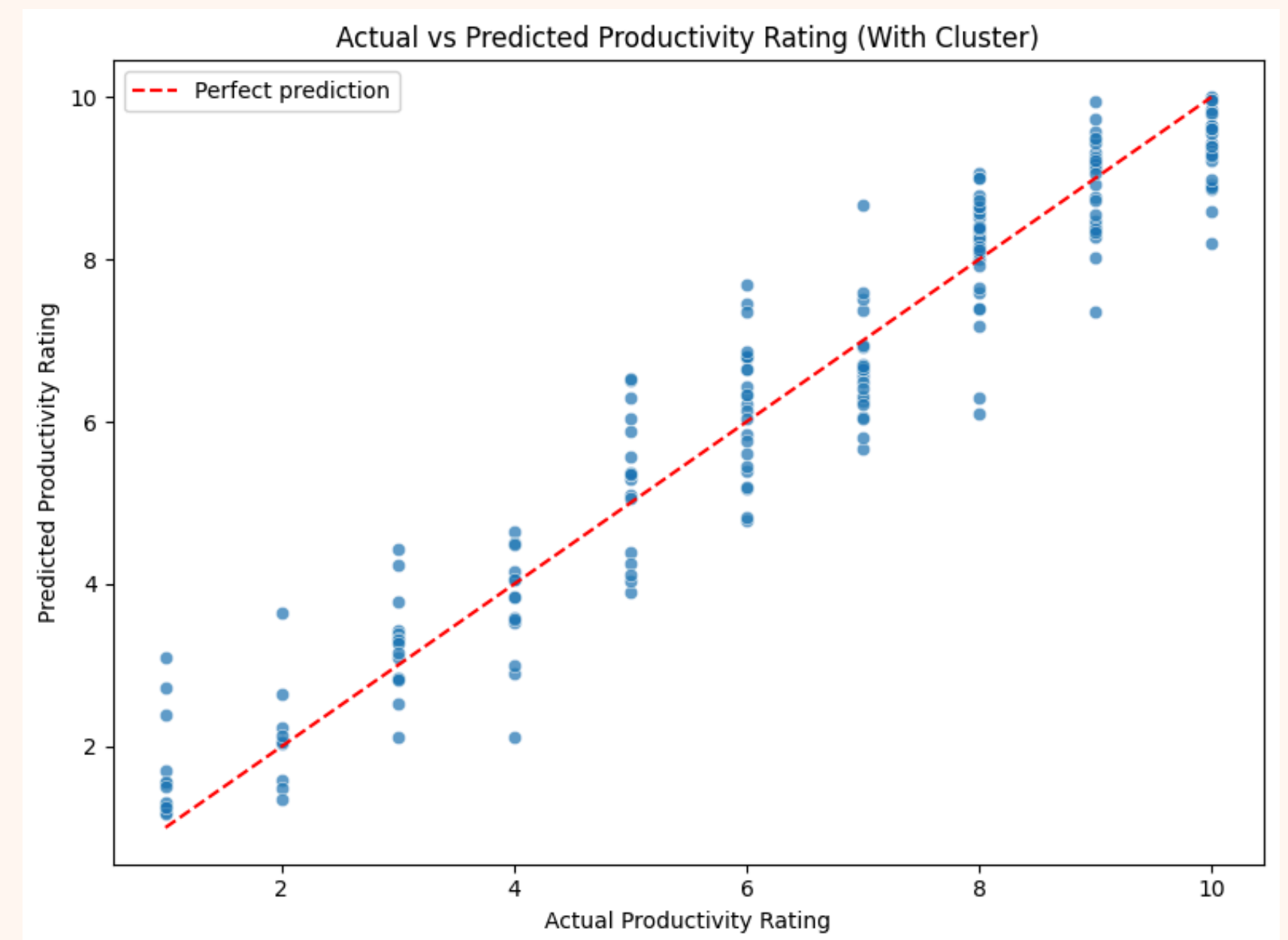
3. Merged Analysis

Predictive Modelling

— Cross-Validated R^2 (5-fold, mean $\pm$ std) —
DecisionTree + Cluster: 0.824 ± 0.041
RandomForest + Cluster: 0.895 ± 0.031

— Holdout Metrics (20% test split) —
DecisionTree + Clust R^2 : 0.831 | MSE: 1.268
RandomForest + Clust R^2 : 0.927 | MSE:
0.544

The cluster-enhanced Random Forest achieved the strongest performance, with holdout $R^2 \approx 0.93$, but this result should be interpreted cautiously because the matching process likely inflated performance through target leakage.



Key Findings

Everyday habits win

Productivity is shaped primarily by daily well-being habits: diet quality, stress, mood and exercise.

Interventions matter little

Nap vs coffee, age, and intervention duration contributed almost nothing to predicting productivity.

Clustering = insight, not lift

K-Means & DBSCAN profile participants well but don't improve prediction beyond raw features.

Merging needs caution

KNN joining is a useful integration tool, but matching on the target inflates apparent accuracy.

Limitations & Takeaways

Limitations:

- Merged $R^2 \approx 0.93$ is not real-world predictive power, it is a by-product of the merge key.
- Dataset 2 carries little predictive signal; results sit at chance level.
- Dataset is synthetic: observed patterns reflect the data generation process rather than real-world relationships. Associations identified in the analysis should not be interpreted as causal.

MAIN TAKEAWAY

Productivity is driven more by day-to-day health and well-being habits than by one-off interventions or background traits.

Thank you!



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